A NONSPLIT SHORT EXACT SEQUENCE OF MODULES THAT BECOMES SPLIT ZARISKI LOCALLY

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file examples.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: The counted proof adds no mathematics of its own; the Zariski-local reading (opens D(p)) is the standard interpretation of the context's 'localize at any prime'.

Editorial note. The label lemma-nonsplit-locally-split names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Context span 1226-1236: the short exact sequence 0 -> M -> direct sum over primes p of Z_(p) -> Q -> 0 where M is the kernel of the map that uses the inclusion Z_(p) -> Q on each summand
- The sequence is nonsplit because there are no nonzero homomorphisms Q -> Z_(p) (a splitting would give a left inverse of the injection Q -> middle term composed to a nonzero map Q -> Z_(p))
- If we localize at any prime p, the sequence becomes split (context closing sentence): the summands with q not equal to p vanish after localization at p

Consider the short exact sequence

$$0 \to M \to \bigoplus\nolimits_{p \text{ prime}} \mathbf{Z}_{(p)} \to \mathbf{Q} \to 0$$

where M is the kernel of the map on the right which uses the inclusion $\mathbf{Z}_{(p)} \to \mathbf{Q}$ on each summand. This sequence of $\mathbf{Z}$-modules is nonsplit because there are no nonzero homomorphisms $\mathbf{Q} \to \mathbf{Z}_{(p)}$. On the other hand, if we localize at any prime p, then the sequence becomes split.

Lemma 0.1. *There exists a ring R and a nonsplit sequence of modules which becomes split Zariski locally.*

Proof. See discussion above. □

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